

Peeyush Khare, Ph.D.

Visit lab group website: [Click here](#) - Khare Research Lab

Research interests: Environmental impacts of anthropogenic activities in megacities; Emerging pollutants in future sustainable cities; Influence of unregulated-burning activities on urban air quality; Development of analytical techniques and automation software for environmental research

Contact Information

104 Center for Environmental Science and Engineering
Kotak School of Sustainability
Indian Institute of Technology (IIT), Kanpur, India

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peeyushkhare90@gmail.com

Academic Appointments

Assistant Professor
(Sept. 2026-current)

Indian Institute of Technology (IIT), Kanpur, India

Kotak School of Sustainability

Research Scientist TVöD E14L4
(June 2024-Aug 2026)

Forschungszentrum Jülich, (Helmholtz-Gemeinschaft) Germany

Institute of Climate and Energy Systems (ICE-3) - Troposphere
Reactive Trace Substances Division

Marie S.-Curie CoFund
Postdoctoral Fellow
(Dec 2020-May 2024)

Paul Scherrer Institute (PSI), Switzerland

PSI Center for Energy and Environmental Sciences
Laboratory of Atmospheric Chemistry

Education

Ph.D. (2020)
Ph.M. (2018)
M.S. (2017)

Yale University, USA

Department of Chemical and Environmental Engineering
Honors in 8 of 9 graduate courses.

PGDip (2015)

Sciences Po Paris France - Ashoka University India (Joint Program)

Environmental Policy (PSIA, SciencesPo) and Liberal Arts (YIF, Ashoka)

M.S. (2014)

Virginia Tech, USA

Department of Civil Engineering (Environmental)

B.Tech. (2012)
(Accredited by ABET, USA)

Vellore Institute of Technology, Vellore India

Department of Civil Engineering
GPA: 9.33/10.0 (Highlight: *Best Outgoing Student*; *Top-10 university-wide*)

Research Theses

Ph.D. (Yale University)

Understanding the Increasing Influence of Non-Combustion Sources on Urban Air Quality.

M.S. (Virginia Tech)

Advisor: Prof. Drew R. Gentner (Chemical and Environmental Engineering)

Simulation of Vertical Conc. Gradient of Influenza Virus Resuspended from Floor Dust

B.Tech. (VIT Vellore)

Advisor: Prof. Linsey Marr (Civil Engineering)

Tackling Failed Monsoons: A Sustainable Approach

Advisor: Prof. Sat Ghosh, FRMetS (Civil Engineering)

Professional Experience Highlights

Forschungszentrum Jülich (June 2024-Aug 2026)

Collaborators: ACTRIS National Facilities, IONICON Analytik Austria, TOFWERK Switzerland, University College Cork Ireland, NOAA USA

- P.I. for Fusion-ToF measurements in international chamber campaigns.
- Ambient measurements using high-resolution Time-of-Flight mass spectrometry.
- Development of ACTRIS chamber measurement guidelines.
- Contributions to ACTRIS PTR-Time-of-Flight measurement guidelines.
- Pan-European audits of ACTRIS measurement sites.
- Advising Ph.D. students and postdocs.

Paul Scherrer Institute (Dec. 2020-May 2024)

Funded by: Swiss Agency for Development and Cooperation (SDC) and the European Commission Horizon 2020 Marie Curie COFUND Grant

Collaborators: TERI, IIT-Kanpur, IIT-Delhi, NEERI, University of Berne, University of Grenoble, Spanish National Research Council.

- Characterization, sources and oxidative potential of ambient PM_{2.5} in India.
- Source apportionment for ambient PM_{2.5} in Barcelona, Spain.
- Developing novel techniques for chemical characterization of organic aerosol.

Supporting contributions to:

- Mobile measurements in Sarajevo, Bosnia using HR-ToF-AMS and Vocus PTR-ToF.
- Offline characterization of PM_{2.5} from megacities in China and Switzerland.
- Development of offline EESI-Orbitrap instrumentation techniques.

Yale University (2015 - 2020)

Funded by: Yale Fellowship & grants from the National Science Foundation (NSF), U.S. Environmental Protection Agency (EPA), NYSEDA and NESCAUM.

Virginia Tech (2012 - 2014)

Funded by: Institute of Critical Technology and Applied Science

Larsen & Toubro, ECC (2010)

VIT Vellore (2009-2012)

Funded by: Villgro Possible and VIT Patent Office

Collaborators: Aerodyne Research, Carnegie Mellon University, U.S. EPA

- Modeling non-combustion emissions in Greater Los Angeles, California.
- Developing techniques to characterize unresolved complex mixture emissions.
- Designing experiments to study environmental impacts of asphalt materials.
- NH₄⁺-ToF field measurements in wintertime New York City on urban emissions.
- Oxidation flow reactor experiments at Aerodyne Research to study asphalt SOA.
- *(solo, PhD-unrelated)*: Ground-water analysis in 6 districts of central India.
- Sorbent tubes sampling campaigns in New York City (2017, 2018).
- Provided I/SVOC emissions data from consumer products to U.S. EPA inventories.
- Experiments to measure turbulent eddies during indoor movement, and;
- FORTRAN-based mathematical modeling to simulate vertical transport of influenza virus indoors resuspended with floor dust during human movement.

Supporting contributions to:

- Vision-based monitoring of the carbon footprint of construction equipment.

Summer Intern, Shrinagar Hydroelectric Power Project, Uttarakhand India

- Training: Reinforced concrete design of the dam, turbine housing & draft tube.
- Fabricated a novel rainfall simulator suitable for hydrological studies in India.
- Stochastic analysis of released droplet trajectories in FORTRAN.
- Received patent from Government of India for the instrument design.

Funded Research Activities

PI: Offline Mass-Spectrometric Analyses of Ambient Particulate Matter Filters from India

Source: The European Commission, Horizon 2020 Marie S. Curie CoFund program

EU Grant Agreement No. 884104

Budget: approx. USD 100,000. (One hundred thousand U.S. Dollars)

Co-PI: Development of Fixed-Wing Drone Platform for Atmospheric VOC Monitoring

At Forschungszentrum Jülich within the ACTRIS CiGas Framework

ACTRIS funding sources: German Federal Ministries (BMBF, BMUV) and institute resources at Forschungszentrum Jülich

Other Grants

- Full Sponsorship, Wilhelm and Else (WE) Heraeus-Foundation Seminar Germany, 2023. (USD 1500)
- Yale Graduate Student Assembly Conference Travel Award, 2019. (USD 750)
- American Geophysical Union: Fall Meeting Travel Grant, 2017. (USD 500)
- NASA Travel Grant for American Association for Aerosol Research Conference, 2017. (USD 500)
- Full Sponsorship, International Students Committee, St. Gallen Symposium, 2016. (USD 2000)
- American Meteorological Society 94th Annual Meeting Travel Grant, 2014. (USD 500)
- Villgro Possible, India- Young Innovator Funding Award, 2011. (USD 500)

Patents

Government of India, The Patent Office. Patent no. 351688 on application: 3623/CHE/2011

Name: *Varun: A Unique Rain Simulator*

Inventors: **Peeyush Khare**, Tushar Agarwal and Satyajit Ghosh

Patentee: Vellore Institute of Technology, VIT University Vellore T.N. India

Purpose: A new rainfall simulator that mimics Indian natural rainfall cost effectively.

Select Awards/ Honors

- **Young Excellent Scientist Program 2025 Awardee** at Forschungszentrum Jülich.
§Nominated among 6 scientists from across all disciplines of the research center.
- **Assistant Professor position** offered at the University of Arizona, USA 2023. (declined)
§Competitive startup package of USD 1.34 million with a tenure track position.
- **Full member, Sigma Xi** - The Scientific Research Honor Society, USA
§Only via nomination.
- **Winning Pitch** at the Hackathon by TUMCREATE and Singapore-ETH Centre at the GYSS 2024, Singapore.
§Proposal: SIMPLE - System for Intelligent Management of Pollutants and Emissions.
- **Atmos. Chem. Colloquium (ACCESS XVI)** program attendee, Brookhaven National Laboratory 2021, USA.
- **Marie S.-Curie CoFund Fellowship** by the European Commission for scientific research at PSI, 2021, Switzerland.
- **St. Gallen Wings of Excellence Award: Leader of Tomorrow**, St. Gallen Symposium 2016, Switzerland.
§Paper chosen in top-21 articles among ~1000 students from 108 nations/350 universities.
- **Young India Fellow** 2014-15.

- **Best Outgoing Student** (School of Mechanical and Building Sciences: Civil+Mech.+Chem. Engineering)
- **Top-10 university-wide in the 2012 graduating batch of ~4000 students** (VIT Vellore), 2012.
- Industrial Endowment Award for excellence in academics, VIT Vellore, 2012.
- Certificate of Distinguished Research Achievements, VIT School of Mechanical and Building Sciences, 2012.
- **Rank 2/72** in the B.Tech Civil Engineering Batch of 2012, VIT Vellore.

Select Academic Scholarships/ Fellowships

- Yale Teaching Fellowships in:

ENAS 335 Professional Ethics, Fall 2018 and 2019.

ENAS 673/ENVE 473 Air Quality and Energy, 2018 and 2019.

PHY 180 Fundamentals of Physics, 2018.

ENAS 120/ENVE 120 Introduction to Environmental Engineering, 2017

- Yale Graduate Fellowship 2015-16.
- Charpak Scholarship (Government of France), 2015.
- Sciences Po Paris full tuition waiver, 2014.
- Balkrishna Industries Tyres Limited (BKT) Scholarship for graduate studies at YIF, 2014.
- Virginia Tech Research and Teaching Assistantships 2012-2014.
- Virginia Tech full tuition waiver for M.S. studies, 2012-2014.
- VIT Annual Merit Scholarships (**All four undergraduate years**), 2008-2012.

Competitive Invitations to Special International Summits/Conferences

- Invited to the **Molecular-Level Understanding of Atmospheric Aerosols (MUOAA) 2024**, Corsica, France.
- Invited to the **Global Young Scientists Summit 2024** by N.R.F, Prime Minister's Office Singapore.
- Delegate at the **4th European Union Clean Air Forum** by the European Commission, Rotterdam 2023.
- Invited to the **782. WE Heraeus Seminar 2023** in Aerosols, Health and Climate: Gigacities and Future, Germany.
- Invited to the **Gordon Research Conference (GRC) in Atmospheric Chemistry 2019**, Newry Maine USA.
- Environmental Scientist delegate at the **World Bank Youth Summit 2018**, W.B. Headquarters Washington DC USA.
- Environmental Scientist delegate at The **Youth Assembly 2018**, NYC USA.

Licenses and Certifications

- *Professional Engineer In-Training (EIT)- Environmental Engineering*

Connecticut Board of Examiners for Professional Engineers and Land Surveyors USA

Verifiable link: <https://account.ncees.org/rn/1825164-1026646-7d786e0>

- *Data Analysis with Python (IBM)*

Coursera (Issued May 2023). Grade: 93%

Credential ID 3A2BJHN838SH

Link to certificate: <https://coursera.org/share/b9f93a83486ddb16d964fc13e425e23b>

Skills: Predictive modeling, Model selection, Data Analysis and Visualization

Laboratory Instrumentation Skills

At the Institute of Climate and Energy Systems (ICE-3): Troposphere, Forschungszentrum Jülich:

- CHARON Fusion-time-of-flight mass spectrometry (IONICON)
- Working with SAPHIR environmental chamber and CACTUS flow tube.
- IoniToF and Ionicon Data Analyzer (IDA) toolkits for ToF data analyses.

At Paul Scherrer Institute, Switzerland:

- High resolution time-of-flight aerosol mass spectrometry (LTOF- and HTOF-AMS)
- Extractive electrospray ionization-Ultra high-resolution orbitrap mass spectrometry (EESI-Orbitrap MS)
- EESI-LTOF MS; EESI-HTOF MS
- In-lab aerosol generation techniques (ESI autosampler and nebulizer)
- Total organic carbon analyzer
- Scanning mobility particle sizer (SMPS)
- Developed analytical methods for EESI-based measurements of non-water-extractable ambient organic aerosol composition

At Gentner Lab, Yale:

- Thermal desorption coupled with gas chromatography electron ionization-mass spectrometry (TD GC EI-MS)
- TD GC-atmospheric pressure chemical ionization-quadrupole-time of flight mass spectrometry (TD GC APCI-Q-TOF)
- Liquid chromatography-electrospray ionization-Q-TOF (LC ESI-Q-TOF)
- Thermal desorption and thermal conditioning techniques for adsorbent quality control.

In collaboration at Aerodyne Research, Billerica MA:

Conducted experiments using potential aerosol mass (PAM) reactor coupled with:

- Time-of flight aerosol mass spectrometer (ToF-AMS) [IGOR toolkits used: Squirrel and Pika]
- VOCUS proton transfer reaction mass spectrometer (VOCUS PTR-TOF) [IGOR toolkits used: Tofware]
- VOCUS ammonium time-of-flight mass spectrometry (VOCUS NH₄⁺-TOF) for urban ambient measurements

Programming Skills: Igor Pro, R, Python, Interactive Data Language, Fortran.

Software Developed

<In Python> *MOLINA*, to automate TOF data analysis for IONICON Fusion-TOF instruments.

<In Igor Pro> *TURACO*, to process PIKA and TOFWARE outputs for PMF in SoFi.

<In Igor Pro> *CHETAK*, to automate the analysis of high resolution GC APCI-Q-TOF data.

Demonstration: Click for Khare video (Auto calibration: 21:00, Auto speciation: 45:00)

Languages : English/Hindi: Native proficiency, French/German/Spanish: Basic/Learner

Invited Talks at:

- [14] Invited Panelist Speaker, Higher Education Series: Panel on Environment, Ashoka University, India, July 2026
- [13] Future of Chemistry Symposium, Dept. of Chemical Sciences, Tata Institute of Fundamental Research, Mumbai 2026
- [12] Environmental Science and Engineering Department, Indian Institute of Technology (IIT), Bombay 2026
- [11] Department of Civil Engineering, Indian Institute of Technology (IIT), Kanpur 2026
- [10] Science Online Talk, Young Excellent Scientist Program, Forschungszentrum Jülich, Germany 2025
- [9] Center for Atmospheric Science, Cambridge University, United Kingdom 2023
- [8] Department of Chemistry, Lehman College, City University of New York, USA 2023
- [7] Distinguished Seminar Series, Department of Civil and Environmental Engineering, Northeastern University, USA 2023
- [6] Department of Chemical and Environmental Engineering, University of Arizona, USA 2023
- [5] School of Marine and Atmospheric Sciences, State University of New York (SUNY) at Stony Brook, USA 2023
- [4] Institute of Energy and Climate Research, Troposphere, Forschungszentrum Jülich, Germany 2022, 2023
- [3] EMPA Laboratory for Air Pollution and Environmental Technology, Switzerland 2022
- [2] Atmospheric Chemistry Department, Leibniz Institute for Tropospheric Research (TROPOS), Germany 2022
- [1] Laboratory of Atmospheric Chemistry, Paul Scherrer Institute, Switzerland 2020

Mentorship Experience

LAC graduate students Paul Scherrer Institute (2021)	Mentees: Jens Top, Marion Gosselin Research: <i>High-resolution chemical speciation of airborne particulate matter from Barcelona, Xi'an (China), Delhi and Lucknow (India).</i>
Teaching Fellow Yale University Fall (2018 and 2019)	ENAS 335 Professional Ethics Course instructor: Prof. Mercedes Carreras Duties: Graded assignments and helped organize guest lectures.
Teaching Fellow Yale University Spring (2018 and 2019)	ENAS 673/ENVE 473: Air Quality and Energy Course instructor: Prof. Drew R. Gentner Duties: Graded assignments.
Mentor Undergraduate Research Yale University (2018)	Mentee: Ricardo Soto Research Title: <i>An Examination of Volatile Organic Compound Emissions from Consumer and Industrial Products</i> Science, Technology and Research Scholars (STARS) Programme
Summer-Term Teaching Yale University (2018)	PHY 180 Fundamentals of Physics Course instructor: Dr. Mehdi Ghiassi-Nejad Duties: Held discussion hours, graded assignments and exams.
Teaching Fellow Yale University Fall (2017)	ENAS 120/ENVE 120: Intro to Environmental Engineering Course instructor: Prof. Jordan Peccia Duties: Held discussion classes and graded assignments.
Graduate Teaching Assistant Virginia Tech (2012-2014)	CEE 4324: Open Channel Flow CEE 5324: Advanced Hydrology CEE 5734: Urban Hydrology Course instructor: Prof. Glenn Moglen Duties: Graded assignments and proctored examinations.

Leadership and Service

ACTRIS auditor	Responsible for auditing ACTRIS CiGas atmospheric measurement stations pan-Europe: • Jungfraujoch research station, Switzerland (November 2024) • PALLAS-Sodankylä research station, Finland (March 2025) • Hyttiälä research station, Finland (March 2025) • SIRTa research station, France (December 2025)
Institute service	• Co-Organizer, ACTRIS CiGas workshops at ACTRIS Weeks. Including: 2024, Italy; 2025, Portugal • 2022, <i>Lead</i>, Source apportionment bi-weekly meetings at the Lab. Atmos. Chem., PSI • 2017, 2018 <i>Member</i>, Graduate Student Recruitment Committee, Yale University • 2018 <i>Member</i>, The Graduate Environmental Engineering Committee, Yale University • 2019, <i>Organizing committee</i>, The Langer Symposium of the Yale Chemical and Environmental Engineering
Non-peer-reviewed publications on mentorship	• <i>Interactive Learning is the Only Way Out</i>. Brainfeed Magazine India, vol. V, issue 5, August 2017. • <i>High School to Engineering College</i>. Brainfeed Magazine India, vol. IV, issue 8, November 2016. • <i>Orienting Young Minds Towards Research and Its Implications for National Prosperity</i>. Brainfeed Magazine India, vol. IV, issue 7, October 2016. • <i>Can Technology Substitute the Teacher?</i> Brainfeed Magazine India, vol. IV, issue 6, September 2016.
Other non-peer-reviewed publications	• <i>Smart Rain Simulators for Rain-Starved Districts of the Indian Subcontinent</i> Khare, P., Agarwal, T., Ghosh, S. Earth and Space Science Open Archive (ESSOAr) https://doi.org/10.1002/essoar.10507296.1. • <i>Transcending Boundaries: My Journey from Engineering to Social Sciences and Back</i>. New Delhi Times, India Published as headlines news in the online edition on August 28, 2019. Link: https://www.newdelhitimes.com/transcending-boundaries-my-journey-from-engineering-to-social-sciences-and-back/ • Contributed a featured article to the book “<i>A Primer for How to Prepare for Overseas Postgraduate Admissions in Engineering</i>”, Macmillan Education Publications India. Authored by: Prof. Sat Ghosh and Ms. Usha Jedusan Affiliated with: Vellore Institute of Technology, VIT University India
Service to academic/scientific community	• Reviewer, Nature Geoscience (Nature); Science Advances (Science); Environmental Science and Technology Letters (ACS); Journal of Geophysical Research: Atmosphere (AGU); Environmental Science: Processes & Impacts Journal (RSC); Atmospheric Chemistry and Physics (EGU); Sustainability Chemistry & Engineering (ACS); Aerosol Measurement Techniques (EGU); Geophysical Research Letters (AGU); Journal of Analytical Atomic Spectrometry (RSC); Atmospheric Science Letters (RMetS); Journal of Materials in Civil Engineering (ASCE); Urban Climate (Elsevier) • Proposals Reviewer for the United States Department of Energy SBIR programs
Public outreach	• <i>The Curious Young Mind: YIF Alumnus’ Journey from Engineering to Liberal Arts to Bringing Environmental Change Worldwide</i>. Interviewed and published in Ashoka University bulletin (and other social media handles) by Shreya Chatterjee (Ashoka University). December 5, 2020 • Interviewed for the special feature section of the Indian national magazine ‘Brainfeed’ in the article titled ‘A Young India Fellow at Yale’, vol. IV, issue 4, July 2, 2016. • Interviewed for the education edition of the Indian national newspaper ‘The Hindu’ in the article titled ‘A Young India Fellow at Yale’ on May 30, 2016. • Interviewed for the education edition of the Indian national newspaper ‘The Times of India’ in the article titled ‘Clear The Air’ on May 23, 2016. • Cover page interview for the Indian national newspaper ‘The Indian Express’s Edex (education edition)’ in the article titled ‘Branch Out Expand Your Boundaries’ on January 12, 2015.

Other Achievements

- Falling Walls Lab, Switzerland 2022. (selected but declined, unforeseen circumstances)
- Invited Delegate at the World Science Forum 2022, South Africa.
- Invited Delegate at the World Business Dialogue 2019, Cologne Germany.
- Jagriti Yatra 2016 India (selected but declined, unforeseen circumstances).
§All India acceptance rate in 2016: ~ 3%

Peer-Reviewed International Journal Publications (Published/submitted: 28, In-prep: 4)

First-author work published [10, incl. corresponding: 2]

[10] *Organic Solvents-Based Offline Aerosol Mass Spectrometry (SOFF-AMS) for Comprehensive Chemical Characterization of Ambient Organic Aerosol*

Khare, P.*, Ioannis, M.M., Dada, L., Kurdieh, A.A., Hao, Y., Diapouli, E., Voiron, C., Jaffrezo, J-L., Styszko, K., Skiba, A., Prevot, A.S.H., Daellenbach, K.R.

In **Environmental Science and Technology (American Chemical Society)**, 59, 34, 18236–18248 (2025) **(I.F. 11.3)**
<https://doi.org/10.1021/acs.est.5c08949>

[9] *Chemical Composition of Fresh and Aged Asphalt-Related Organic Aerosols: From Ambient Observations to Laboratory Experiments*

Khare, P., Machesky, J., Williams, L., Humes, M., Canagaratna, M., Krechmer, J.E., Lambe, A.T., Presto, A.A., Gentner, D.R.

In **Environmental Science and Technology | Air (American Chemical Society)**, 2, 4, 446-455 (2025) **(I.F. pending)**
<https://doi.org/10.1021/acsestair.4c00193>

[8] *Detailed Chemical Characterization and Source Apportionment of Coarse Airborne Particulate Matter in Urban and Rural Catalonia, Spain*

joint-first-[in ‘t Veld, M., Khare, P.], Hao, Y., Reche, C., Pérez, N., Alastuey, A., Yus-Díez, J., Marchand, N., Prevot, A.S.H., Querol, X., Daellenbach, K.R.

In **Science of the Total Environment (Elsevier)**, 902, 166440 (2023) **(I.F. 8.0)**
<https://doi.org/10.1016/j.scitotenv.2023.166440>

[7] *Ammonium-adduct Chemical Ionization to Investigate Anthropogenic Oxygenated Gas-Phase Organic Compounds in Urban Air*

Khare, P., Krechmer, J.E., Machesky, J. E., Hass-Mitchell, T., Cao, C., Wang, J., Majluf, F., Lopez-Hilfiker, F., Seltzer, K., Pye, H.O.T., Commane, R., McDonald, B., Toledo-Crow, R., Mak, J.E., Gentner, D.R.

In **Atmospheric Chemistry and Physics (European Geosciences Union)**, 22, pp. 14377-14399 (2022) **(I.F. 5.1)**
<https://doi.org/10.5194/acp-22-14377-2022>

[6] *Asphalt-related Emissions are a Major Missing Non-Traditional Source of Secondary Organic Aerosol Precursors*

Khare, P., Machesky, J., Soto, R., He, M., Presto, A., Gentner, D.R.

In **Science Advances (Science/AAAS)**, 6(36), eabb9785 (2020) **(I.F. 12.5)**
<https://doi.org/10.1126/sciadv.abb9785>

This publication was covered by many popular media outlets worldwide: The Guardian, CNBC, NewScientist, Associated Press, MSN News, ABC News, Smithsonian Magazine, Phys.org among others.

Science| Research Highlight: “It’s not just cars that make pollution. It’s the roads they drive on, too.” September 3, 2020.

[5] *Considering the Future of Anthropogenic Gas-Phase Organic Compound Emissions and the Increasing Influence of Non-Combustion Sources on Urban Air Quality.*

Khare, P., Gentner, D.R.

In **Atmospheric Chemistry and Physics (European Geosciences Union)**, 18, pp. 5391-5413 (2018) **(I.F. 5.1)**
<https://doi.org/10.5194/acp-18-5391-2018>

Selected as highlight article by the journal.

[4] *Advances in Offline Approaches for Trace Measurements of Complex Organic Compound Mixtures via Soft Ionization and High-Resolution Tandem Mass Spectrometry*

Khare, P., Marcotte, A., Sheu, R., Walsh, A., Ditto, J., Gentner, D.R.

In **Journal of Chromatography A (Elsevier)**, 1598, pp. 163-174 (2019) **(I.F. 4.0)**
<https://doi.org/10.1016/j.chroma.2019.03.037>

[3] *A Large Scale Investigation of the Quality of Groundwater in Six Major Districts of Central India during the 2010-2011 Sampling Campaign.*

Khare, P.*

In **Environmental Monitoring and Assessment (Springer International)**, 189:429. (2017) **(I.F. 3.0)**
<https://doi.org/10.1007/s10661-017-6130-0>

[2] *Simulation of Vertical Concentration Gradient of Influenza Viruses in Dust Resuspended by Walking.*

Khare, P., Marr, L.C.

In **Indoor Air (Wiley)**. Vol. 25, 428-440. (2014) **(I.F. 4.3)**
<https://doi.org/10.1111/ina.12156>

[1] *Droplet Spectrum Analysis from Artificially Generated Rain Showers.*

Khare, P., Agarwal, T., Ghosh, S.

In **ASCE Journal of Irrigation and Drainage Engineering**, Vol. 140(8), ISSN (online): 1943-4774. (2014) (**I.F. 2.1**)

[https://doi.org/10.1061/\(ASCE\)IR.1943-4774.0000742](https://doi.org/10.1061/(ASCE)IR.1943-4774.0000742)

Co-authored papers published/submitted [18]

[18] *Assessing Organic Aerosol Sources in Barcelona via Non-Targeted Molecular Characterization*

Arianna Tronconi et al

(submitted)

[17] *A Decadal-Scale Perspective on PM₁₀ Composition and its Variability Drivers at the High-Alpine research station Jungfraujoch*

Julian Weng et al.

(submitted)

[16] *Recreating the Breath of Cities in the Atmospheric Simulation Chamber SAPHIR*

Wu et al.

(submitted)

[15] *Regional and Urban Drivers of PM_{2.5} Oxidative Potential Across China*

Cui et al.

(submitted)

[14] *Multi-Stress Interaction Effects on BVOC emission Fingerprints in Oak and Beech: A Cross-Investigation Using Neural Networks and Positive Matrix Factorization*

Biplob Dey, Toke Due Sjøgren, **Peeyush Khare**, Yizhen Wu, Sindhu Vasireddy, Georgios Gkatzelis, Martin Schultz, Alexander Knohl, Riikka Rinnan, Thorsten Hohaus, Eva Y. Pfannerstill

In **Biogeosciences (European Geosciences Union)** 23(4) 1423-1457 (2026) (**I.F. 3.9**)

<https://doi.org/10.5194/egusphere-2025-3779>

<https://doi.org/10.5194/bg-23-1423-2026>

[13] *Real-Time Source Apportionment Approaches in Six Chinese Cities Using the AXA (ACSM, Xact, Aethalometer) Instrumental Set-Up*

Manousos I. Manousakas, Tianqu Cui, Qiyuan Wang, Lu Qi, Cheung Ka Yuen, Lubna Dada, Yufang Hao, **Peeyush Khare**, Tian Jie, Yuemei Han, Jay G. Slowik, Junji Cao, Kaspar R. Daellenbach, André S. H. Prévôt

In **Scientific Reports (Springer Nature)** 16 (9890) (2026) (**I.F. 3.9**)

<https://doi.org/10.1038/s41598-026-38154-x>

[12] *Characterization of Coarse Particulate Matter in Urban & Rural Switzerland using Advanced Offline Mass Spectrometry*
Ortiz Beltran K.S., Cui, T., Casotto, R., Lamkaddam, H., Tobler, A., Hao, Y., **Khare, P.**, Manousakas, M., Dada, L., Grange, S.K., Huglin, C., Uzu, G., Jaffrezo, J-L., Rausch, J., Jaramillo-Vogel, D., El-Haddad, I., Slowik, J.G., Prevot, A.S.H., Daellenbach, K.R.

In **Atmosphere (MDPI)** 17 (199) (2026) (**I.F. 2.3**)

<https://doi.org/10.3390/atmos17020199>

[11] *Assessing the Severe Urban Pollution Crisis in Sarajevo, Bosnia and Herzegovina: Mobile Measurements and Source Characterization*

Michael Bauer, Jay G. Slowik, Marta Via, **Peeyush Khare**, Benjamin Chazeaua, Kristina Glojek, Manousos Manousakas, Zachary C.J. Decker, Asta Gregorič, Almir Bijedić, Enis Krečinić, Griša Močnik, Katja Džepina, André S.H. Prévôt

In **Environment International (Elsevier)** 207 (110009) (2026) (**I.F. 9.7**)

<https://doi.org/10.1016/j.envint.2025.110009>

This publication was covered by several popular media outlets worldwide: The Guardian, Phys.Org among others.

[10] *Transported Smoke from Crop Residue Burning as the Major Source of Organic Aerosol and Health Risks in Northern Indian Cities During Post-Monsoon*

Hao, Y., Straehl, J., **Khare, P.**, Cui, T., Schneider-Beltran, K.S., Qi, L., Wang, D., Top, J., Surdu, M., Bhattu, D., Bhowmik, H.S., Vats, P., Rai, P., Kumar, V., Ganguly, D., Szidat, S., Uzu, G., Jaffrezo, J-L., Rastogi, N., Slowik, J., Haddad, I.E., Tripathi, S.N., Prevot, A.S.H., Daellenbach, K.R.

In **Environment International (Elsevier)** 202 (109583) (2025) (**I.F. 9.7**)

<https://doi.org/10.1016/j.envint.2025.109583>

This publication was covered by many popular media outlets worldwide: The Guardian, The Indian Express among others.

[9] *Measurement Report: Sources and Meteorology Influencing Highly-Time Resolved PM_{2.5} Trace Elements at 3 Urban Sites in Extremely Polluted Indo-Gangetic Plain in India*

Shukla, A.K., Tripathi, S.N., Talukdar, S., Murari, V., Gaddamidi, S., Manousakas, M-I., Lalchandani, V., Dixit, K., Ruge, V.M., **Khare, P.**, Kumar, M., Singh, V., Rastogi, N., Tiwari, S., Srivastava, A.K., Ganguly, D., Daellenbach, K.R., Prevot, A.S.H.

In **Atmospheric Chemistry and Physics (European Geosciences Union)** 25 (6), pp. 3765–3784 (2025) (**I.F. 5.1**)
<https://doi.org/10.5194/acp-25-3765-2025>

[8] *Composition and Sources of Organic Aerosol in Two Megacities in Western China Using Multiple Mass Spectrometric and Statistical Techniques*

Cui, T., Manousakas, M.I., Wang, Q., Uzu, G., Hao, Y., Huang, X., **Khare, P.**, Qi, L., Han, Y., Slowik, J.G., Jaffrezo, J.-L., Cao, J., Prevot, A.S.H., Daellenbach, K.R.

In **Environmental Science and Technology | Air (American Chemical Society)**, 1 (9), pp. 1053–1065 (2024)
<https://doi.org/10.1021/acsestair.4c00051>

Received ACS ES&T Air best paper award for the year 2024.

[7] *Local Incomplete Combustion Emissions Define the PM_{2.5} Oxidative Potential in Northern India*

Bhattu, D., Tripathi, S.N., Bhowmik, H.S., Moschos, V., Lee, C.P., Rauber, M., Salazar, G., Abbaszade, G., Cui, T., Slowik, J.G., Vats, P., Mishra, S., Lalchandani, V., Satish, R., Rai, P., Casotto, R., Tobler, A., Kumar, V., Hao, Y., Qi, L., **Khare, P.**, Manousakas, M.I., Wang, Q., Han, Y., Tian, J., Rastogi, N., Srivastava, A.K., Ganguly, D., Bjelic, S., Canonaco, F., Schnelle-Kreis, J., Dominutti, P.A., Jaffrezo, J.-L., Szidat, S., Chen, Y., Cao, J., Baltensperger, U., Uzu, G., Daellenbach, K.R., El Haddad, I., Prévôt, A.S.H.

In **Nature Communications (Nature)**, 5 (3517), (2024) (**I.F. 15.7**)

<https://doi.org/10.1038/s41467-024-47785-5>

This publication was covered by many popular media outlets worldwide: MSN News, Times of India, Phys.org among others.

[6] *Total Organic Carbon Measurements Reveal Major Gaps in Petrochemical Emissions Reporting*

He, M., Ditto, J., Gardner, L., Machesky, J., Hass-Mitchell, T.N., Chen, C., **Khare, P.**, Sahin, B., Fortner, J.D., Plata, D.L., Drollette, B.D., Hayden, K.L., Wentzell, J.J.B., Mittermeier, R.L., Leithead, A., Lee, P., Darlington, A., Wren, S.N., Zhang, J., Wolde, M., Moussa, S.G., Li, S.-M., Liggio, J., Gentner, D.R.

In **Science (AAAS)**, 383(6681), pp. 426–432 (2024) (**I.F. 45.8**)

<https://doi.org/10.1126/science.adj6233>

This publication was covered by many popular media outlets worldwide: The Guardian, The Independent, CBC News, MSN News, Phys.org, Nature among others.

[5] *Anthropogenic Secondary Organic Aerosol and Ozone from Asphalt-Related Emissions*

Seltzer, K.M., Rao, V., Pye, H.O.T., Murphy, B.N., Place, B.K., **Khare, P.**, Gentner, D.R., Allen, C., Cooley, D., Mason, R., Houyoux M.

In **Environmental Science: Atmospheres (Royal Society of Chemistry)** (**I.F. 3.5**)

<https://doi.org/10.1039/D3EA00066D>

[4] *Effects of Organic Aerosol Molec.-Level Compositional Variability on Phase State and Thermodynamic Mixing Behavior*

Ditto, J.C., Joo, T., **Khare, P.**, Bui, A.A.T., Griffin, R.J., Ng, N.L., Gentner, D.R.

In **Environmental Science and Technology (American Chemical Society)**, 53(22), pp. 13009–13018 (**I.F. 11.3**)

<https://doi.org/10.1021/acs.est.9b02664>

[3] *Advances in Offline Approaches for Speciated Measurements of Trace Gas-Phase Organic Compounds via an Integrated Sampling-to-Analysis System*

Sheu, R., Marcotte, A., **Khare, P.**, Charan S., Ditto, J., Gentner, D.R.

In **Journal of Chromatography A (Elsevier)**, 1575, pp. 80–90 (2018) (**I.F. 4.0**)

<https://doi.org/10.1016/j.chroma.2018.09.014>

[2] *An Omnipresent Diversity and Variability in the Chemical Composition of Atmospheric Functionalized Organic Aerosol*

Ditto, J.C., Barnes, E., **Khare, P.**, Takeuchi, M., Joo, T., Bui, A.A.T., Lee-Taylor, J., Eris, G., Chen, Y., Aumont, B., Jimenez, J.L., Ng, N.L., Griffin, R.J., Gentner, D.R.

In **Communications Chemistry (Nature)**, 1:75 (2018) (**I.F. 6.2**)

<https://doi.org/10.1038/s42004-018-0074-3>

One of the Editor's Picks of the Year for anniversary collection in Communications Chemistry (Nature).

[1] *How Custom-Built Rain Simulators Counter the Effects of Failed Monsoons: An Agricultural and Hydrological Study.*

Agarwal, T., **Khare, P.**, Ghosh, S.

In OIDA **International Journal of Sustainable Development**, Vol. 4(1), pp. 35–42. (2012)

Published Datasets

[2] <https://doi.org/10.6084/m9.figshare.21507798> - Ambient concentration timeseries dataset for non-traditional emissions in New York City. Associated paper: Ammonium adduct chemical ionization to investigate anthropogenic oxygenated gas-phase organic compounds in urban air

[1] [10.5281/zenodo.16880946](https://doi.org/10.5281/zenodo.16880946) - Mass spectral datasets for newly developed offline aerosol mass spectrometry technique. Associated paper: Organic Solvents-Based Offline Aerosol Mass Spectrometry (SOFF-AMS) for Comprehensive Chemical Characterization of Ambient Organic Aerosol

Contributions in International Conferences/Meetings/Workshops

Corresponding author/presenter contributions [29 of 85 total]

Oral presentations [18]

- [18] *Molecular Drivers of Organic Aerosol in Ambient Reactive Mixtures from Emerging Urban Sources Simulated in the SAPHIR Chamber*
International Aerosol Conference 2026, Xi'an, China
(accepted)
- [17] *Implementation Highlights from the Contributing Units to the in-situ Reactive Trace Gases (CiGas) Measurements in the ACTRIS Research Infrastructure Framework*
ACTRIS Week 2025, Evora, Portugal
- [16] *Detailed Molecular Characterization of Limonene Secondary Organic Aerosol Under Varying Oxidation Conditions and Reaction Timescales at the SAPHIR Chamber during CHANEL campaign*
European Geophysical Union (EGU) General Assembly 2025, Vienna, Austria
- [15] *Perspectives on External Quality Assessment of the VOC Data Reported by ACTRIS Measurement Platforms - Current Methods and Looking Forward*
ACTRIS Data QA/QC Online Workshop 2025 (pan-Europe meeting)
- [14] *Molecular-Level Characterization of Secondary Organic Aerosol from Limonene Emissions Under Varying Oxidation Conditions and Reaction Timescales in the SAPHIR Chamber during CHANEL campaign*
Helmholtz Topic Days 2024: Changing Earth - Sustaining our Future, Bad Schönborn, Germany
- [13] *Development of Solvents-Based Offline Aerosol Mass Spectrometry for Comprehensive Chemical Characterization of Ambient Organic Aerosol*
European Aerosol Conference 2024, Tampere Finland
- [12] *Sources and Oxidation Potential of Fine Airborne Particulate Matter in Megacities of North and South India*
Molecular-Level Understanding of Atmospheric Aerosols (MUOAA) Meeting 2024, Corsica, France
- [11] *Application of Offline Aerosol Mass Spectrometry for Identification and Characterization of Sources of Airborne Particulate Matter in Lucknow and Pune and their Oxidative Potential*
The Swiss Agency for Development and Cooperation (SDC)-India International Collaboration Meeting 2024, Villigen, Aargau Switzerland
- [10] *Molecular-Level Chemical Characterization and Source Apportionment of Ambient Water-Soluble Organic Aerosol in Megacities of North and South India*
Source Apportionment Workshop 2023, Indian Institute of Technology (IIT)-Bombay, Mumbai, India
- [9] *Molecular Characterization of Organic Aerosols in Two Heavily Polluted Cities of India: Sources and Seasonality*
Indian Aerosol Science and Technology Association (IASTA) Conference 2023, Mumbai India
- [8] *Ambient Observations and Laboratory Characterization of Asphalt-Related Secondary Organic Aerosol*
International Global Atmospheric Chemistry (iCACGP-IGAC)-Early Career Researchers Conference 2023
- [7] *Offline Characterization and Source Apportionment of Water-Soluble Organic Fraction of PM_{2.5} from Indian Megacities*
Datalystica SoFi Pro Users Meeting 2023, Switzerland
- [6] *Molecular-level Chemical Characterization and Source Apportionment of Ambient Water-Soluble Organic Aerosol in Two Megacities of North and South India*
European Aerosol Conference 2023, Malaga Spain
- [5] *Laboratory Investigations of Asphalt-Related Emissions - A Major Nontraditional Source of Secondary Organic Aerosol Precursors*
16th Atmospheric Chemistry Colloquium for Emerging Senior Scientists (ACCESS XVI), Brookhaven Nat. Lab., 2021 USA
- [4] *Modeling the Dispersion of Resuspended Virus Laden Floor Dust due to Walking in an Indoor Environment.*
Virginia Tech Fluid Mechanics Symposium 2013, USA.
- [3] *A Novel and Economical Technique to Determine the Carbon Dioxide Emissions from the Construction Equipment-Vision Based Monitoring and Control of Construction Equipment Carbon Footprint.*
Environmental Protection Agency EPA-Expo 2013, Washington DC, USA.
- [2] *A Novel Rain Simulator for Rain Starved Districts in India.*
Young Innovators Programme, Indian Institute of Technology (IIT), Madras India
- [1] *How Custom Built Rain Simulators Counter the Effect of Failed Monsoons-An Agricultural and Hydrological Study.*
International Conference on Sustainable Development 2011, Malaysia.

Poster presentations [11]

- [11] *Demonstration of the Interoperability of Two Atmospheric Simulation Chambers*
ACTRIS Science Conference 2026, Oslo, Norway
- [10] *Constraining the Molecular-Level Chemical Composition and Sources of Air Pollution in Indian Megacities*
Global Young Scientists Summit 2024, National University of Singapore, Singapore

- [9] *Understanding the Sources of the Organic Composition of Coarse Airborne Particulate Matter (PM₁₀) in Urban and Rural Catalonia, Spain*
European Aerosol Conference 2023, Malaga Spain
- [8] *Elucidating Variations in Molecular-Level Speciation and Sources of Ambient Organic Aerosol in Two Megacities of North and South India*
782. Wilhelm und Else Heraeus-Foundation Seminar 2023 on Aerosols, Health and Climate: Gigacity and Future, Bad Honnef, Germany
- [7] *Detailed Chemical Speciation and Source Apportionment of PM_{2.5} in Densely Populated Indo-Gangetic Region Using High-Resolution Mass Spectrometry*
International Aerosol Conference (IAC), 2022 Athens, Greece
- [6] *Secondary Organic Aerosol Formation from Gas-Phase Asphalt-Related Emissions in a Potential Aerosol Mass Reactor.*
American Geophysical Union (2019), San Francisco USA
- [5] *Emissions of Complex Gas-Phase Organic Compound Mixtures from the Application and Use of Asphalt Materials Under Typical Environmental Stresses and Emissions Timescales*
Gordon Research Conference 2019: Atmospheric Chemistry, Newry ME USA
- [4] *An Investigation of Non-Combustion-Related Emissions of Gas-Phase Organic Compounds in Multiple Major U.S. Cities*
American Association for Aerosol Research (AAAR) - 10th International Aerosol Conference 2018, St. Louis MO USA
- [3] *Next Generation Offline Approaches to Trace Organic Compound Speciation: Approaching Comprehensive Speciation with Soft Ionization and Very High Resolution Tandem Mass Spectrometry.*
American Geophysical Union (AGU) 2017, New Orleans LA USA
- [2] *Evolving Urban Emissions: Increasing Impact of Non-Combustion Secondary Organic Aerosol Precursor Sources on Urban Air Quality.*
American Association for Aerosol Research Conference (AAAR) 2017, Raleigh NC USA
- [1] *Stop Measuring Happiness from Material Gains.*
Leaders of Tomorrow Summit, St. Gallen Symposium 2016, Switzerland.

Co-authored contributions [56 of 85 total]

Please click here for a detailed list: [Peeyush Khare - list of publications](#)

Select Non-Technical Skills

Black Belt in Shotokan Karate (Conferred by Japan Karate Association)
Assistant Instructor in Krav Maga

References

Upon request

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